

Game 01 — Tap or Don't Tap (Complete Plan)

Game 1 of 7 Complexity: ★ Low Estimate: 1–2 days Status: not started

Complete plan (supersedes the 2026-09-09 sample). Status: **built & browser-verified** (2026-09-09 build pass) — `apps/frontend/public/games/tap-or-dont-tap/` with decoys, Stroop traps, rule swap, audio, pause and a baked percentile table. Logic tests: `apps/frontend/src/__tests__/games-tap-or-dont-tap.test.ts`.

This doc is the spec of record: the constants below are authoritative when tuning, and the open items in §9–§12 are the only remaining work.

1. Overview

A Go/No-Go reaction game (the real psychology self-control test). The shareable number is the **best reaction time in ms**. Success metric: median first session ≥ 15 rounds; players can state their ms score from memory after one session.

2. Complete game spec

Round loop (state machine: `MENU → WAITING → SIGNAL → FEEDBACK → (WAITING | GAMEOVER)` , plus `PAUSED`):

- WAITING** — dark screen, random delay **uniform 300–3000 ms**. A tap here is a **false start**: lose a heart, skip the round (no signal shown).
- SIGNAL** — one of:
 - GREEN → tap within the window.
 - RED → do not tap; survives after the window expires.
 - ● / ● DECOY (round ≥ 10) → never tap; ignoring it for the full window = +50.
 - STROOP TRAP (round ≥ 15) — word "TAP" in red or "WAIT" in green; **color wins, word ignored**.
- FEEDBACK** — 400 ms overlay ("218ms! 🔥" / "❌ Too slow" / "✅ Resisted +25"), then 250 ms blank before the next WAITING.

Constants (authoritative — implemented in `game.js`)

Constant	Value	Notes
<code>WAIT_MIN/MAX_MS</code>	300 / 3000	uniform random
<code>WINDOW_START_MS</code>	1200	signal visible duration, round 1
<code>WINDOW_SHRINK_MS</code>	25	per round
<code>WINDOW_FLOOR_MS</code>	450	never below
<code>HEARTS</code>	3	0 → game over
<code>GREEN_BASE_POINTS</code>	100	<code>points = min(100 + streak×10, 500)</code> (×5 cap)

Constant	Value	Notes
RESIST_BONUS	25	red expired untouched
DECOY_BONUS	50	decoy expired untouched (round $\geq$ 10)
FEEDBACK_MS	400	+ 250 ms blank
SWAP_ROUNDS	21–23	"🔄 RULES SWAPPED!" — red taps, green doesn't

Signal mix per round band: 1–9 → 45 % green / 55 % red; 10–14 → 40/30/30 decoy;
15+ → 40 green / 30 red / 15 decoy / 15 Stroop.

Losing: tap on red/decoy/Stroop-wrong, false start, or green missed → –1 heart each.
Reaction recorded only on green hits. Swap inverts green/red only; decoys always "never tap".

3. Progression & percentile model

- Difficulty = shrinking window + signal mix. No other knobs.
- **Best reaction** = min green-hit reaction of the run; **score** = sum of points.
- **Top-% badge**: `BAKED_TABLE` in `game.js` maps score → percentile (fixed simulated distribution). Personal history (last 50 runs `{date, score, bestMs}`) stored locally.

4. Screens & UI

State	Elements
menu	Title, one-line rule, best score + best ms, ▶ Start, mute toggle
waiting	Full-dark screen, "wait for it..." hint, HUD (hearts, score, round)
signal	Full-bleed color, optional overlay word (Stroop), HUD persists
feedback	Center overlay text + tone color, 400 ms
gameover	Card: score, best ms, rounds, top-% badge, 📄 Share, ↺ Retry, ☰ Menu
paused	Overlay "Paused — tap to resume" (auto on <code>visibilitychange</code>)

Whole screen is the tap target (`pointerdown`); `touch-action:none` on the play area;
Space/Enter equivalent on desktop; HUD height fixed across states (no relayout).

5. Data model (localStorage, guarded wrapper)

```
game:tap-or-dont-tap:best      → { score: 1240, bestMs: 187 }
game:tap-or-dont-tap:history  → [{ date: '2026-09-10', score: 1240, bestMs: 187 }, ...max 50]
game:tap-or-dont-tap:muted    → true|false
```

6. File structure & function inventory

```
tap-or-dont-tap/  
  index.html    # state sections + gameover card  
  style.css     # flash colors, HUD, overlays  
  game.js      # constants → pure core → state machine → DOM wiring
```

Pure core (no DOM — the test surface, currently in `game.js`, mirrored in the site test

file): `windowMs(round)`, `pointsForGreen(streak)`, `pickSignal(round, rng)`,
`resolveTap({signalKind, expectsTap, tapped, elapsedMs})`, `percentileFor(score)`.

DOM layer: `startRun`, `beginRound`, `showSignal`, `handleTap`, `expireSignal`,
`finishRound`, `loseHeart`, `gameOver`, `pauseRun`, `showFeedback`, audio helpers.

7. Edge cases (verify each once)

1. Hidden tab during WAITING → pause; on return restart the wait from 0. Hidden during SIGNAL → counts as a miss (anti-cheat).
2. Tap within 50 ms of a state change → single tap only (pointerdown debounce).
3. Private mode / storage disabled → game runs, persistence silently skipped.
4. Phone sleep mid-wait → `visibilitychange` → pause path.
5. Stroop × swap overlap (rounds 21–23) → swap inverts green/red; decoys never tappable.
6. Rapid restart from gameover → all pending timers cleared.
7. Reaction ≤ 0 ms or > window + 50 ms → discarded as anomaly (clock sanity).

8. Testing plan

Existing: `apps/frontend/src/__tests__/games-tap-or-dont-tap.test.ts` — keep green.

Required cases (verify present; add missing): `windowMs` ramp incl. floor;

`pointsForGreen` cap; `resolveTap` truth table (5 signal kinds × tap/late/no-tap);

seeded `pickSignal` sequences per band; `percentileFor` monotonic + bounds.

9. Task breakdown

P0 — playable core — DONE (2026-09-09 build pass)

- ☒ State machine + round loop; signal kinds; hearts; reaction capture; ramp; scoring
- ☒ HUD + gameover card; pause/resume (`visibilitychange` incl. wait-restart)

P1 — full rules & feel — DONE

- ☒ Streak multiplier; decoys; Stroop traps; rule swap; false starts
- ☒ Feedback overlays; WebAudio + mute; baked percentile badge

P2 — persistence/share/QA

- ☒ Best score + history persistence; share chain
- ☐ **Delete** `analyticsCandidates` / `trackGamePlayed` **stubs** (isolation decision — they must never post to `/api/*`)
- ☐ Optional: migrate inline storage/audio helpers to `../shared/`
- ☐ Optional: extract pure core to `core.js`; keep site test file green
- ☐ QA gate (master README §5): offline play + isolation greps

P3 — polish

- ☐ Reduced-motion variant of flash; tuning pass after 5 test sessions
- ☐ History sparkline on menu; haptics on heart loss; sound design pass

10. Acceptance criteria

- §7 edge cases each verified once (note date when done); §8 tests green.
- 20 consecutive rounds, no stuck state; restart from gameover < 300 ms to first signal.
- Share works through all three fallback paths (share sheet / clipboard / prompt).

11. Deferred coupling (intentionally NOT built)

Footer/nav links, `game_played` analytics posting, achievements, leaderboard backend — per master README §7. The analytics stubs in `game.js` are to be **deleted**, not wired.

12. Remaining work (from code review 2026-09-10)

1. Remove analytics stubs (P2 — the only required code change).
2. Optional shared-utils migration + `core.js` extraction (P2, non-blocking).
3. Reduced-motion + tuning (P3).